

FLEXIBLE CIRCUITS GETTING BETTER

Circuits that stretch, fold or twist will improve numerous devices. One particular application is biomedical sensors that can be comfortably implanted in the human body

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Electronics circuits till now are fabricated mostly on hard substrates. This rigidity imposes shape limitations for electronic equipment. Though miniaturisation of integrated circuits has helped to a large extent in giving the desired shape to electronic equipment, much more can be achieved with the latest developments in the fabrication of flexible circuits.

Flexible circuitry reduces the size and weight of a finished product. It allows increased circuit density and eliminates bulky connections and wiring. And the added ability to fold the circuit expands the boundaries of design and packaging.

The underlying technology

Flexible electronics, also known as flex circuits, or flex circuit

boards, is a technology for assembling electronic circuits by mounting electronic devices on flexible high-performance plastic substrates, such as polyimide. Circuits can be designed in configurations from simple, single-sided conductive paths to complex, high-density, three-dimensional assemblies utilising a variety of fabrication materials from low-cost polyester to mid-range PEN to high-density adhesiveless polyimide Novaclad. Additionally, flex circuits can be screen-printed silver circuits on polyester.

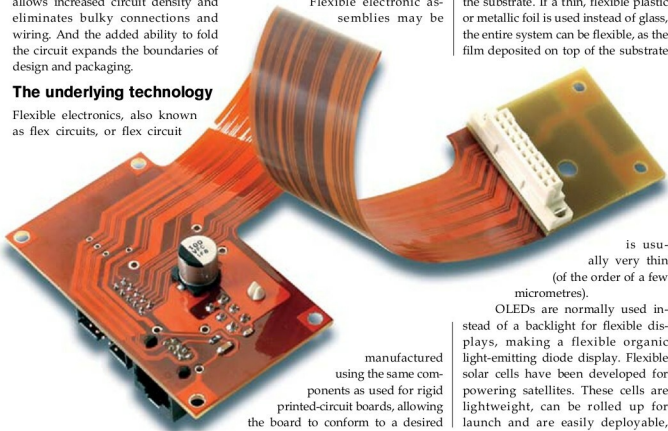
Flexible electronic assemblies may be

shape, or to flex during its use.

Developments so far

The original method for flexible CMOS circuits comprised a 2- to 3-micrometre circuit layer sitting atop a plastic substrate as much as 100µm thick. It could curve around a small roll of coins. But the new version has a total thickness of only 1.7 µm, including the plastic, which gives it the ability to wrap around a rod whose diameter is roughly 85 µm.

In LCD fabrication, glass is used as the substrate. If a thin, flexible plastic or metallic foil is used instead of glass, the entire system can be flexible, as the film deposited on top of the substrate



is usually very thin (of the order of a few micrometres).

OLEDs are normally used instead of a backlight for flexible displays, making a flexible organic light-emitting diode display. Flexible solar cells have been developed for powering satellites. These cells are lightweight, can be rolled up for launch and are easily deployable,

manufactured using the same components as used for rigid printed-circuit boards, allowing the board to conform to a desired